Project 4 | Exercises 1 and 2

Pendulum swing-up and car navigation with RRT and KPIECE1

Both systems use the assigned equations and OMPL control planners. The pendulum starts at (-pi/2, 0) and targets (pi/2, 0), with angular velocity limited to +/-10 and torque bounds 3, 5, and 10. RRT and KPIECE1 use the same physical bounds, start/goal query, and goal tolerance for each system.

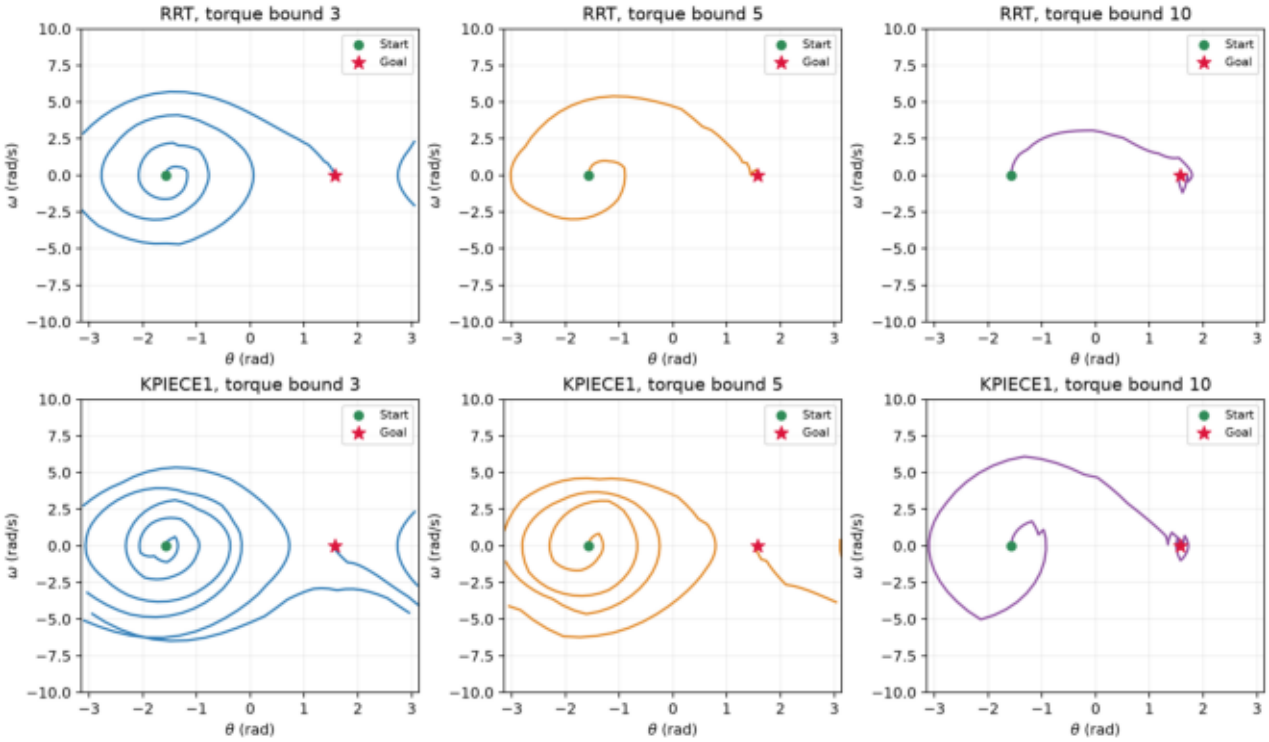
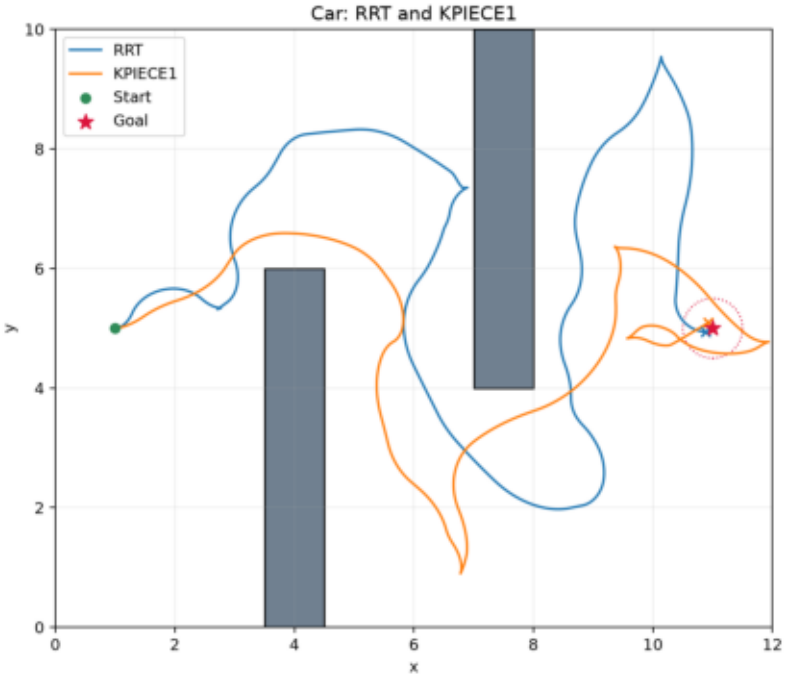


Figure 1. Actual pendulum trajectories; periodic angle seams are broken. Green: start. Red: goal.



Environment and results

Figure 2. The point car moves in a 12 x 10 workspace. Alternating rectangular barriers (3.5,0,1,6) and (7,4,1,6) block the straight start-goal route. Both trajectories pass above the first barrier and below the second.

The car starts at (1,5,0,0) and targets (11,5,0,0). Velocity lies in [-2,2], allowing reverse motion. Angular velocity and acceleration each lie in [-1.5,1.5]. Crosses mark final positions; the dotted circle is the spatial envelope of the full goal region.

Low pendulum torque requires repeated swings to build energy. In these seed-1 examples, RRT motion durations are 8.25, 4.75, and 3.00 s for torque bounds 3, 5, and 10. KPIECE1 durations are 13.25, 9.00, and 4.40 s. These are feasible examples, not optimized paths.

Planning setup and validation

The goal radius is 0.1 for the pendulum and 0.5 for the car in their compound state-space metrics.
Integration step: 5 ms; propagation step: 50 ms. All eight configurations passed three seeded trials.
Independent control replay checked dynamics, bounds, collisions, and goal arrival. Exercises 3 and 4 remain unimplemented.